

$$= (BC + BA \cdot 1) + AC' \text{ (Postulate 2(b))}$$

$$= (BC + BA \cdot (AC')) + AC' \text{ (Postulate 5(a))}$$

$$= (BC + (BAC + BAC')) + AC' \text{ (Postulate 4(a))}$$

$$= ((BC + BAC) + BAC') + AC' \text{ (Theorem 4(a))}$$

$$= ((BC + B \cdot (AC)) + BAC') + AC' \text{ (Theorem 4(b))}$$

$$= ((BC + B \cdot (CA)) + BAC') + AC' \text{ (Postulate 3(b))}$$

$$= (BC + BCA) + BAC' + AC' \text{ (Theorem 4(b))}$$

$$= (BC \cdot 1 + BCA) + BAC' + AC' \text{ (Postulate 2(b))}$$

$$= (BC \cdot (1+A)) + BAC' + AC' \text{ (Postulate 4(a))}$$

$$= (BC \cdot (A+1)) + BAC' + AC' \text{ (Postulate 3(a))}$$

$$= (BC \cdot 1 + BAC') + AC' \text{ (Theorem 2(a))}$$

$$= (BC + BAC') + AC' \text{ (Postulate 2(b))}$$

$$= BC + (BAC' + AC') \text{ (Theorem 4(a))}$$

$$= BC + (B \cdot (AC') + AC') \text{ (Theorem 4(b))}$$

$$= BC + (AC') \cdot (B + AC') \text{ (Postulate 3(b))}$$

$$= BC + ((AC') \cdot (B + AC' \cdot 1)) \text{ (Postulate 2(b))}$$

$$= BC + ((AC') \cdot (B + 1)) \text{ (Postulate 4(a))}$$

$$= BC + AC' \cdot 1 \text{ (Theorem 2(a))}$$

$$= BC + AC' \text{ (Postulate 2(b))}$$

$$(c) [(CD)' + A]' + A + CD + AB \text{ to three literals.}$$

$$= ((CD)')' \cdot A' + A + CD + AB \text{ (Theorem 5(a))}$$

$$= CD \cdot A' + A + CD + AB \text{ (Theorem 3)}$$

$$= CD \cdot A' + (A + CD + AB) \text{ (Theorem 4(a))}$$

$$= CD \cdot A' + (AB + (A + CD)) \text{ (Postulate 3(a))}$$

$$= CD \cdot A' + ((AB + A) + CD) \text{ (Theorem 4(a))}$$

$$= CD \cdot A' + ((AB + A \cdot 1) + CD) \text{ (Postulate 2(b))}$$

$$= CD \cdot A' + ((A \cdot (B + 1)) + CD) \text{ (Postulate 4(a))}$$

$$= CD \cdot A' + (A \cdot (B + 1) + CD) \text{ (Theorem 2(a))}$$

$$= CD \cdot A' + (A + CD) \text{ (Postulate 2(b))}$$

$$= CD \cdot A' + (CD + A) \text{ (Postulate 3(a))}$$

$$= (CD \cdot A + CD) + A \text{ (Theorem 4(a))}$$